

概率论第一次自学作业

Caratheodary extension theorem

Assume that $\mathcal{E}$ is a semi-algebra on Ω , $\mu(\cdot) : \mathcal{E} \rightarrow [0, +\infty)$ is a *additive function*, i.e.

- $\mu(\emptyset) = 0$
- $\forall A_n \in \mathcal{E}$, if $\sum_{n=1}^{\infty} A_n \in \mathcal{E}$, then

$$\mu\left(\sum_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

then: 1) μ has an extension measure $\bar{\mu}$ on $(\Omega, \sigma(\mathcal{E}))$, s.t. $\bar{\mu}|_{\mathcal{E}} = \mu$;

2) If μ is σ -finite function (on $\mathcal{E}$), i.e. there exist $\{A_n\}$ s.t.

$$\Omega = \sum_{n=1}^{\infty} A_n, A_n \in \mathcal{E}, \mu(A_n) < +\infty, \forall n \geq 1$$

then $\bar{\mu}$ is unique.

Lemma 1

Define

$$\mu^*(A) := \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : A_n \in \mathcal{E}, A \subset \bigcup_n A_n \right\}, \forall A \subset \Omega$$

then: i) $\mu^*(\emptyset) = 0$;

ii) $A_1 \subset A_2 \implies \mu^*(A_1) \leq \mu^*(A_2)$;

iii) $\mu^*\left(\bigcup_n A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n), \forall A_n \in \Omega$;

iv) $\mu^*|_{\mathcal{E}} = \mu$.

Proof: The properties i) and ii) are trivial. For iii), let $\varepsilon > 0$, for each j , by the definition of $\mu^*(A_j)$, there exists a covering $\{B_{j_k}\}$ of A_j such that

$$\sum_{k=1}^{\infty} \mu(B_{j_k}) \leq \mu^*(A_j) + \frac{\varepsilon}{2^j}$$

The double sequence $\{b_{j,k}\}$ is a covering of $\bigcup_j A_j$ such that

$$\sum_j \sum_k \mu(B_{j,k}) \leq \sum_j \mu^*(A_j) + \varepsilon \implies \mu^*\left(\bigcup_j A_j\right) \leq \sum_j \mu^*(A_j) + \varepsilon, \forall \varepsilon > 0$$

Let $\varepsilon \rightarrow 0$, then we get property iii).

For iv), let $A \in \mathcal{E}$, then the single A serves as a covering of A , hence $\mu^*(A) \leq \mu(A)$.

For any covering $\{B_j\}$ of A , we have $AB_j \in \mathcal{E}$ and

$$\bigcup_j AB_j = A \in \mathcal{E}$$

From property iii), we have

$$\mu(A) = \mu\left(\bigcup_j AB_j\right) \leq \sum_j \mu(AB_j) \leq \sum_j \mu(B_j) \implies \mu(A) \leq \mu^*(A)$$

So we get $\mu^*|_{\mathcal{E}} = \mu$.

Remark: In the above proof, we have used:

$$\mu\left(\bigcup_j A_j\right) \leq \sum_j \mu(A_j), \forall \{A_j\} \subset \mathcal{E} \wedge \bigcup_j A_j \in \mathcal{E}$$

It's easy to see that if we denote $B_1 = A_1, B_n = A_1^c \cdots A_{n-1}^c A_n, n \geq 2$, then $\{B_n\} \subset \mathcal{E}$ and $\{B_n\}$ are a sequence of disjoint sets. Thus

$$\mu\left(\bigcup_j A_j\right) = \mu\left(\sum_j B_j\right) = \sum_j \mu(B_j) \leq \sum_j \mu(A_j)$$

It's trivial that for all $A_1, A_2 \in \mathcal{E}, A_1 \subset A_2$, we have $\mu(A_1) \leq \mu(A_2)$.

Lemma 2

Define

$$\mathcal{F}^* := \{A \subset \Omega : \mu^*(Z) = \mu^*(ZA) + \mu^*(ZA^c), \forall Z \subset \Omega\} \quad (1)$$

Then: i) $\mathcal{F}^*$ is a σ -algebra on Ω ;

ii) $\mathcal{E} \subset \mathcal{F}^* (\implies \sigma(\mathcal{E}) \subset \mathcal{F}^*)$;

iii) μ^* is a measure on $\mathcal{F}^*$.

Proof: Let $A \in \mathcal{E}$. For any $Z \subset \Omega$ and $\varepsilon > 0$, there exists a covering $\{B_j\}$ of Z s.t.

$$\sum_j \mu(B_j) \leq \mu^*(Z) + \varepsilon$$

Since $AB_j \in \mathcal{E}, \{AB_j\}$ is a covering of $AZ; \{A^c B_j\}$ is a covering of $A^c Z$; hence

$$\mu^*(AZ) \leq \sum_j \mu(AB_j); \mu^*(A^c Z) \leq \sum_j \mu(A^c B_j)$$

Since μ is a measure on $\mathcal{E}$, we have for each j :

$$\mu(AB_j) + \mu(A^c B_j) = \mu(B_j)$$

So we have

$$\mu^*(AZ) + \mu^*(A^c Z) \leq \sum_j \mu(B_j) \leq \mu^*(Z) + \varepsilon$$

Letting $\varepsilon \rightarrow 0$ establishes the criterion (1) in its $\geq$ form. The $\leq$ form is trivial, thus we know $A \in \mathcal{F}^*$, which means $\mathcal{E} \subset \mathcal{F}^*$.

To prove that $\mathcal{F}^*$ is a σ -algebra, it is trivial that it is closed under complementation because the criterion (1) is unaltered when A is changed into A^c . Next, to show that $\mathcal{F}^*$ is closed under union, let $A, B \in \mathcal{F}^*$. Then for any $Z \subset \Omega$, from (1), we have

$$\begin{aligned}\mu^*(ZA) &= \mu^*(ZAB) + \mu^*(ZAB^c) \\ \mu^*(ZA^c) &= \mu^*(ZA^c B) + \mu^*(ZA^c B^c)\end{aligned}$$

Hence by (1) again as written:

$$\mu^*(Z) = \mu^*(ZAB) + \mu^*(ZAB^c) + \mu^*(ZA^c B) + \mu^*(ZA^c B^c)$$

Applying (1) with Z replaced by $Z(A \cup B)$, we have

$$\begin{aligned}\mu^*(Z(A \cup B)) &= \mu^*(Z(A \cup B)A) + \mu^*(Z(A \cup B)A^c) \\ &= \mu^*(ZA) + \mu^*(ZA^c B) \\ &= \mu^*(ZAB) + \mu^*(ZAB^c) + \mu^*(ZA^c B)\end{aligned}$$

Comparing the two preceding equations, we see that

$$\mu^*(Z) = \mu^*(Z(A \cup B)) + \mu^*(Z(A \cup B)^c)$$

Hence $A \cup B \in \mathcal{F}^*$, and we have proved that $\mathcal{F}^*$ is an algebra.

Now let $\{A_j\}$ be an infinite sequence of sets in $\mathcal{F}^*$; put

$$B_1 = A_1, B_j = A_1^c \cdots A_{j-1}^c A_j, j \geq 2$$

Then $\{B_j\}$ is a sequence of disjoint sets in $\mathcal{F}^*$ and has the same union as $\{A_j\}$. For any $Z \subset \Omega$, we have for each $n \geq 1$:

$$\begin{aligned}\mu^*\left(Z \bigcup_{j=1}^n B_j\right) &= \mu^*\left(Z \left(\bigcup_{j=1}^n B_j\right) B_n\right) + \mu^*\left(Z \left(\bigcup_{j=1}^n B_j\right) B_n^c\right) \\ &= \mu^*(Z B_n) + \mu^*\left(Z \left(\bigcup_{j=1}^{n-1} B_j\right)\right)\end{aligned}$$

Because $\{B_n\} \subset \mathcal{F}^*$, it follows by induction on n that

$$\mu^* \left(Z \bigcup_{j=1}^n B_j \right) = \sum_{j=1}^n \mu^*(ZB_j) \quad (2)$$

Since $\bigcup_{j=1}^n B_j \in \mathcal{F}^*$ and the monotonicity of μ^* , we have:

$$\begin{aligned} \mu^*(Z) &= \mu^* \left(Z \bigcup_{j=1}^n B_j \right) + \mu^* \left(Z \left(\bigcup_{j=1}^n B_j \right)^c \right) \\ &\geq \sum_{j=1}^n \mu^*(ZB_j) + \mu^* \left(Z \left(\bigcup_{j=1}^{\infty} B_j \right)^c \right) \end{aligned}$$

Letting $n \rightarrow \infty$ and using property iii) in *Lemma 1*, we obtain

$$\mu^*(Z) \geq \mu^* \left(Z \bigcup_{j=1}^{\infty} B_j \right) + \mu^* \left(Z \left(\bigcup_{j=1}^{\infty} B_j \right)^c \right)$$

which establishes $\bigcup_{j=1}^{\infty} B_j \in \mathcal{F}^*$. Thus $\mathcal{F}^*$ is a σ -algebra.

Finally, let $\{B_j\}$ be a sequence of disjoint sets in $\mathcal{F}^*$. By the property ii) in *Lemma 1* of μ^* and (2) with $Z = \Omega$, we have

$$\begin{aligned} \mu^* \left(\sum_{j=1}^{\infty} B_j \right) &\geq \mu^* \left(\sum_{j=1}^n B_j \right) \\ \Rightarrow \mu^* \left(\sum_{j=1}^{\infty} B_j \right) &\geq \overline{\lim}_{n \rightarrow \infty} \mu^* \left(\sum_{j=1}^n B_j \right) = \overline{\lim}_{n \rightarrow \infty} \mu^* \left(\bigcup_{j=1}^n B_j \right) = \sum_{j=1}^{\infty} \mu^*(B_j) \end{aligned}$$

Combined with the property iii) of μ^* , we obtain the countable additivity of μ^* on $\mathcal{F}^*$, i.e. for any disjoint sequence $\{B_j\} \subset \mathcal{F}^*$, we have

$$\mu^* \left(\bigcup_{j=1}^{\infty} B_j \right) = \sum_{j=1}^{\infty} \mu^*(B_j)$$

This establishes that μ^* is a measure on $\mathcal{F}^*$.

Lemma 3

If μ_1 and μ_2 are two σ -finite measure on $\sigma(\mathcal{E})$, $\mathcal{E}$ is an algebra, $\mu_1|_{\mathcal{E}} = \mu_2|_{\mathcal{E}}$, then $\mu_1 = \mu_2$.

Proof: Let $\{\Omega_n\} \subset \mathcal{E}$ satisfies $\Omega = \sum_{n=1}^{\infty} \Omega_n$, $\mu_1(\Omega_n) < +\infty$, $\forall n \in \mathbb{N}$.

Then $\mu_2(\Omega_n) < +\infty, \forall n$. Define a class $\mathcal{C}$ of subsets of Ω as follows:

$$\mathcal{C} = \{A \subset \Omega : \mu_1(\Omega_n A) = \mu_2(\Omega_n A), \forall n \in \mathbb{N}\}$$

From $\mu_1|_{\mathcal{E}} = \mu_2|_{\mathcal{E}}$, it's trivial that $\mathcal{E} \subset \mathcal{C}$. Suppose $A_k \in \mathcal{C}, A_k \subset A_{k+1}, \forall k \in \mathbb{N}$ and $A_k \uparrow A$. Then by the upper continuity of measure, we have

$$\mu_1(\Omega_n A) = \lim_k \mu_1(\Omega_n A_k) = \lim_k \mu_2(\Omega_n A_k) = \mu_2(\Omega_n A), \forall n \in \mathbb{N}$$

Thus $A \in \mathcal{C}$. Similarly, from the lower continuity of measure and

$$\mu_1(\Omega_n) = \mu_2(\Omega_n) < +\infty, \forall n \in \mathbb{N}$$

If $A_k \in \mathcal{C}$ and $A_k \downarrow A$, then $A \in \mathcal{C}$. Therefore μ is closed under both increasing and decreasing limits, which means $\mathcal{C}$ is a M.C. From the MCT, we have

$$\mathcal{E} \subset \mathcal{C} \implies \sigma(\mathcal{E}) = m(\mathcal{E}) \subset \mathcal{C}$$

This implies that for any $A \in \sigma(\mathcal{E})$, from the upper continuity, we have

$$\mu_1(A) = \mu_1\left(A \bigcup_{n=1}^{\infty} \Omega_n\right) = \lim_{n \rightarrow \infty} \mu_1\left(A \bigcup_{i=1}^n \Omega_n\right) = \lim_{n \rightarrow \infty} \mu_2\left(A \bigcup_{i=1}^n \Omega_n\right) = \mu_2(A)$$

Thus $\mu_1|_{\sigma(\mathcal{E})} = \mu_2|_{\sigma(\mathcal{E})}$, i.e. $\mu_1 = \mu_2$.

Proof of the extension theorem

Existence: From *Lemma 1*, we know that μ can be extended to an outer measure μ^* on Ω with $\mu^*|_{\mathcal{E}} = \mu$. From *Lemma 2*, μ^* is a measure on $\mathcal{F}^* \supset \sigma(\mathcal{E})$. Define

$$\bar{\mu} = \mu^*|_{\sigma(\mathcal{E})}$$

Then $\bar{\mu}$ is a measure on $\sigma(\mathcal{E})$ which satisfies $\bar{\mu}|_{\mathcal{E}} = \mu$. Thus $\bar{\mu}$ is an extension measure on $(\Omega, \sigma(\mathcal{E}))$ of μ .

Uniqueness: If μ_1 and μ_2 are two σ -finite functions on $\sigma(\mathcal{E})$ and $\mu_1|_{\mathcal{E}} = \mu_2|_{\mathcal{E}}$. From *Lemma 3*, we have $\mu_1 = \mu_2$.

Q.E.D.